

When is anti-holomorphic dependence forced?

Exchange non-Hermiticity of one-way transfer kernels and a proposed time-domain test in a recirculating fiber loop

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Abstract

We investigate when the anti-holomorphic (conjugate-variable) dependence of a complex field or kernel is forced by the underlying physics, as opposed to being removable by a choice of gauge, basis, or preparation. As a mathematical result, we show that for two counter-winding mode ladders coupled unidirectionally — via reservoir-engineered, Lindblad-type coupling with a Jordan-block (double-pole) structure per mode pair and a linear frequency ladder — the two-point transfer kernel sums to a closed form involving the Lerch transcendent in the product of conjugated coordinates. Exchange-Hermiticity of the kernel and physical reciprocity are two independent invariants: the reciprocity content is carried by the exchange invariant $R = K - S(K)$ alone, evaluated on the complete two-point transfer kernel, and exchange-Hermitian kernels exist that are not reciprocal (and conversely), so the mirror partner certifies neither property by itself. The one-way kernel violates both invariants: unidirectionality deletes the mirror partner and, independently, makes $R \neq 0$. All symbolic claims are certified by exact Wirtinger-derivative computation. We then propose an implementation in a recirculating fiber loop with an intracavity amplifier and an in-loop isolator, together with a time-domain protocol based on a secular revival envelope, a revival comb, and a single-versus-twin-comb conditional reciprocity witness; a direct measurement of the reciprocity invariant additionally requires reverse injection with phase-referenced detection, specified but not simulated here. Numerical simulations with literature-typical component values indicate that the secular exponent remains robust under several realistic noise sources, while revival-comb visibility is significantly more demanding in terms of loop-phase stability. Beyond the protocol, we characterize the kernel's intrinsic analyticity boundary at $|x| = 1$ — emergent from the mode ladder, with a chiral phase law of its branch-cut discontinuity separating the one-way kernel from its real-offset (scalar) control — and prove an elementary Fourier floor theorem for the interference relief along the cut: the residual relief of large kernel sums is the window-contrast of the Fourier transform of the weighted winding measure, with closed-form floors for the canonical ladder families and an amplitude dichotomy (the relief of a finite ladder is Bohr almost periodic and cannot decay; the amplitude of an absolutely continuous continuum dies away — the persistence of the interference minima being a separate property, exemplified by the uniform sinc whose exact zeros recur forever). Read backwards, the law yields a spectroscopy of the surviving transfer element, recovering its winding measure from the relief alone, with proven limits. No experimental measurements have been performed; experimental verification remains necessary.

1 Introduction

1.1 The question

A complex field or kernel on a two-dimensional carrier generally depends on both the holomorphic coordinate z and its conjugate $\bar{z}$. In many physical settings the anti-holomorphic dependence is an artifact of description: it can be removed by a gauge transformation, by a change of basis, or by a

different choice of preparation or probe, after which the object is holomorphic, or real, or a holomorphic function dressed by a modulus factor. This paper is concerned with the opposite situation: when is the anti-holomorphic dependence *forced* — present in every gauge, every basis, and for every admissible preparation? We treat removability operationally: a structure is removable if some transformation within a declared admissible class (unitary or similarity gauge transformations, basis changes among the physical modes, choices of drive or input port) maps the object to one whose mixed Wirtinger structure is trivial. Forcedness is the failure of every such attempt, and we require it to be certified symbolically rather than argued.

1.2 Why one-point fields are insufficient

Two elementary mechanisms empty the one-point case, and motivate the two-point object studied here. First, reality locking: any physically real scalar observable satisfies a conjugation constraint that ties its anti-holomorphic content to its holomorphic content point by point; the two are mirror images and carry no independent information. Second, preparation erasability: even for a genuinely complex field obtained by driving a system, a one-point snapshot of the driven response depends on the drive, and there exist admissible drives that erase its anti-holomorphic content entirely; structure that can be switched off by an input choice is not forced. Both mechanisms leave a natural escape: an object that is neither a real observable nor a driven snapshot, namely the two-point transfer kernel of an open, non-reciprocal system — the response at one point to a source at another. A kernel is a property of the dynamical generator, not of any drive; and one-way coupling, implemented at the level of the generator (Lindblad-type), deletes the conjugate-paired sector that reality locking enforces and, independently, breaks the reciprocity invariant $R = K - S(K)$ on the complete transfer kernel (Section 2.3). The body of the paper makes this escape quantitative.

1.3 Results, and what is not claimed

We establish, at the level of symbolically certified formula structure: (i) a closed form for the two-point transfer kernel of two counter-winding mode ladders under per-pair one-way Jordan coupling — a Lerch transcendent in the product of conjugated coordinates; (ii) an independence theorem — exchange-Hermiticity ($\Delta = K - M(S(K))$) and reciprocity ($R = K - S(K)$) are independent axes, all four zero/non-zero combinations being realized by certified kernels, and one-way coupling violates both, deleting the mirror partner and making $R \neq 0$ on the complete transfer kernel; (iii) a three-fold classification of the resulting kernel (conjugate-sector entanglement, non-pairing, transcendence) with the paired-sum constructions as certified reciprocal counterexamples and the reproducing-kernel (Bergman) form as the certified Hermitian decoy ($\Delta = 0, R \neq 0$). At the level of machine-verified simulation and one elementary hand proof, we further establish: (iv) the kernel’s intrinsic, emergent analyticity boundary and the chiral phase law of its cut, with class-level genericity; (v) a Fourier floor theorem for the interference relief of large kernel sums, with sampling-dependent convergence rates and an amplitude dichotomy for the continuum limit (decay of amplitude, not oblivion of the minima); (vi) closed-form floors for the canonical ladder families; (vii) a spectroscopy of the surviving transfer element, with proven limits. We further propose a fiber-loop implementation and a time-domain protocol, and we analyze its robustness against literature-typical imperfections. We claim no experimental result: no measurement has been performed, all component values are catalogue-typical, and the relation of the kernel to superficially similar arithmetic objects (mock modular forms) is discussed and explicitly not asserted. The contribution, if any, is a sharply posed question, a certified structural dichotomy, an elementary machine-faced floor theorem with closed-form consequences, an inverse instrument whose limits are proven, and a concrete, falsifiable measurement proposal.

Beyond the forcing question itself, the proposed protocol bears on an open experimental debate in exceptional-point (EP) sensing. The claimed sensitivity enhancement of EP-based sensors has been challenged on noise grounds [1, 2], and fundamental-limit analyses have shown that non-reciprocity —

rather than EP proximity per se — is the resource that allows a sensor to exceed the bounds constraining any reciprocal device [3]. Both strands of that debate presuppose the ability to certify, in operation and under realistic noise, (i) that a device is genuinely defective (a Jordan-block degeneracy rather than a near-degenerate diagonalizable pair) and (ii) that it is genuinely non-reciprocal. The time-domain observables proposed here address precisely these two certification tasks at low cost: the secular exponent of the revival envelope is a direct witness of Jordan structure which our simulations indicate is robust to loop-phase jitter and between-shot noise (Section 8), and the single-versus-twin revival comb is, within the loop model, a conditional witness of broken reciprocity. We emphasize that these are certification tools; no sensitivity-enhancement claim is made or implied.

1.4 Outline

Section 2 constructs the kernel and states the dichotomy. Section 3 characterizes the kernel’s analyticity boundary: its intrinsic and emergent character, the chiral phase law of its cut, the interference relief along the tear, and the log-scale calendar of its deep minima. Section 4 states and proves the Fourier floor theorem, establishes the amplitude dichotomy of the continuum limit — and its boundary: amplitude decay does not erase the minima — and computes the canonical floors in closed form. Section 5 reads the law backwards into a spectroscopy of the surviving transfer element, with proven limits. Section 6 situates the object among its nearest neighbors. Section 7 describes the proposed platform and protocol; Section 8 the imperfection analysis. Section 9 discusses assumptions, limitations, open questions, and falsifiability. The certification details, the full proof of the floor theorem, the closed-form derivations, and the simulation parameters are provided as supplementary material in the repository accompanying this deposit.

2 Mathematical structure of the one-way kernel

2.1 Setup

We consider two families of modes with opposite winding on a ring-like carrier: an “upstream” family $\phi_m \propto e^{+im\theta}$ and a “downstream” family $\chi_m \propto e^{-im\theta}$, indexed by $m \geq 1$. In complex coordinates $z = re^{i\theta}$ on (or near) the carrier circle, the upstream family is holomorphic-like (z^m up to a radial profile) and the downstream family anti-holomorphic-like ($\bar{z}^m$). Each pair (ϕ_m, χ_m) is coupled unidirectionally: the downstream mode is driven by the upstream one with amplitude g_m , while the reverse matrix element vanishes identically. Such one-way coupling can be produced by reservoir engineering in the sense of cascaded quantum systems [4, 5, 6]: the vanishing of the reverse element is then a property of the Lindblad generator itself, not of a drive or preparation choice, which is the property we rely on throughout. When the two modes of a pair are degenerate and damped at equal rates — which is automatic if they are frequency-degenerate partners coupled to the same bus — the per-pair dynamical matrix is triangular with equal diagonal entries, i.e., a 2×2 Jordan block: the transfer function from upstream to downstream carries a double pole,

$$G_m(\omega) = \frac{g_m}{(i(\omega - \omega_m) - \kappa/2)^2}. \quad (1)$$

We take the pair frequencies on a linear ladder, $\omega_m = m\delta$ (free spectral range δ), with a common linewidth κ .

2.2 Closed form of the two-point transfer kernel

The object of interest is not a one-point field but the two-point transfer kernel: the response at observation point $\mathbf{r}_1$ to a localized source at point $\mathbf{r}_2$. Writing the mode sums at fixed probe frequency

and collecting the winding bookkeeping (downstream factor $\bar{z}_1^m$ at the observation point, conjugated upstream factor $\bar{w}_2^m$ at the source point), the kernel takes the form

$$G_{21}(\mathbf{r}_1; \mathbf{r}_2) = \sum_{m \geq 1} g_m x^m, \quad x = \bar{z}_1 \bar{w}_2, \quad (2)$$

convergent for $|x| < 1$ (observation and source inside the carrier circle). We write G_{21} for this surviving transfer element; the complete two-channel kernel is the matrix $\mathbf{G} = (G_{ij})$ with $G_{12} = 0$, and for matrix-valued kernels the exchange S also transposes the channel indices, $(S\mathbf{G})_{ij}(\mathbf{r}_1, \mathbf{r}_2) = G_{ji}(\mathbf{r}_2, \mathbf{r}_1)$, or explicitly in its C -covariant form $(S_C \mathbf{G})(\mathbf{r}_1, \mathbf{r}_2) = C^{-1} \mathbf{G}(\mathbf{r}_2, \mathbf{r}_1)^T C$: with this definition the scalar element, a function of the commuting product $\bar{z}\bar{w}$, is exchange-symmetric on its own, while the matrix kernel has $R_{\mathbf{G}} = \mathbf{G} - S(\mathbf{G}) \neq 0$. In the general statements of this paper K denotes a generic two-point kernel — scalar or matrix, with S acting accordingly — and we write G_{21} or $\mathbf{G}$ whenever the distinction matters; reciprocity statements always refer to the matrix (Section 2.3). The weight ladder determines the analytic class:

- (a) flat weights $g_m = g$ give the geometric (algebraic) kernel $gx/(1-x)$;
- (b) weights g/m give the logarithmic kernel $-g \log(1-x)$;
- (c) the physical Jordan ladder $g_m = k/\lambda_m^2$ with $\lambda_m = -\kappa/2 - im\delta$ gives

$$G_{21} = -\frac{k}{\delta^2} \left[\Phi(x, 2, c) - \frac{1}{c^2} \right], \quad c = \frac{\kappa}{2i\delta}, \quad (3)$$

where Φ is the Lerch transcendent; the leading structure is a dilogarithm.

Each closed form was verified against its defining series by partial-sum comparison (forty terms, several sample points, geometric tail bound); we found direct high-order numerical differentiation of the Lerch function unstable and avoided it. We emphasize that case (a) shows the algebraic and transcendental classes are independent of the one-way construction: flat weights give an *algebraic* surviving element G_{21} , and it is the physical double-pole ladder that supplies the transcendental class — one-wayness itself lives in $G_{12} = 0$ with $G_{21} \neq 0$ on the matrix kernel, never in the scalar closed form (Section 2.3).

Version note (v11). This version repairs the reciprocity statement of the published v3 (DOI 10.5281/zenodo.21317960). Exchange-Hermiticity and reciprocity, stated as equivalent in v3 (abstract, contribution (ii), and this section), are independent invariants; the Bergman reproducing kernel, used in v3 as a reciprocal control, is exchange-Hermitian but not reciprocal ($\Delta = 0, R \neq 0$) and is reclassified below as the Hermitian decoy; the reciprocity invariant must be evaluated on the complete two-point transfer kernel, the scalar closed form being exchange-symmetric by itself; and the cut-phase witness is stated comparatively, with its sufficiency-only limit (Section 3.2). No closed form, simulation number, or figure changed; the measurement protocol was strengthened with the reverse-injection acquisition and the phase-referenced detection that a direct measurement of $R_{\mathbf{G}}$ requires. The repair was prompted by an external adversarial audit and is certified by an expanded repair harness built in the same exact symbolic discipline as the original claims. Intermediate drafts v4–v10 existed only internally and were never deposited.

2.3 Reciprocity and exchange-Hermiticity: two independent invariants

Let S denote the exchange of the two points, $(z, \bar{z}) \leftrightarrow (w, \bar{w})$ (for matrix-valued kernels S is understood in the port basis, $C = I$, so that $S \equiv S_C$; a general calibration enters through the S_C of Section 2.2, and the invariant is then written $R_{\mathbf{G}, C} = \mathbf{G} - S_C(\mathbf{G})$), and let the mirror map M act on a kernel formula by swapping each variable with its conjugate together with $i \rightarrow -i$. We call a kernel *exchange-Hermitian* when $K = M(S(K))$ up to an additive constant; because logarithmic branches can contribute a piecewise

constant, the operationally safe test is the constancy of $\exp(K - M(S(K)))$. Alongside $\Delta = K - M(S(K))$ we define the *reciprocity invariant* $R = K - S(K)$: reciprocity is the statement $R = 0$ (the response from 2 to 1 equals the response from 1 to 2), while $\Delta = 0$ is an anti-linear Hermiticity condition. These two invariants are *independent* — conflating them was the error of earlier versions of this paper — and all four zero/non-zero combinations are realized by certified kernels:

quadrant	certified representative	invariants
reciprocal–Hermitian	$g(z, \bar{z}) + g(w, \bar{w})$, real coefficient, $M(g) = g$	$R = 0, \Delta = 0$
dissipative reciprocal	$c(g(z, \bar{z}) + g(w, \bar{w}))$, $c \notin \mathbb{R}$, $M(g) = g$	$R = 0, \Delta \neq 0$
Hermitian decoy	Bergman $-k \log(1 - z\bar{w})$, $k \in \mathbb{R}$	$R \neq 0, \Delta = 0$
one-way	matrix kernel, $G_{21} = f(\bar{z}\bar{w})$, $G_{12} = 0$	$R \neq 0, \Delta \neq 0$

The dissipative-reciprocal row requires the stated hypothesis: the complex coefficient breaks Δ only if the building block is itself mirror-fixed ($M(g) = g$, $g \neq 0$); dissipation is fully compatible with reciprocity, a point long emphasized in the electromagnetic literature [22, 23]. At the operator level, with a symmetric non-degenerate pairing C and the reciprocal transpose $A^\sharp = C^{-1}A^T C$ (no complex conjugation, hence loss-compatible), $L = L^\sharp$ for the generator is equivalent to $\mathbf{G} = \mathbf{G}^\sharp$ for the matrix kernel $\mathbf{G} = L^{-1}$, i.e. to $R_{\mathbf{G},C} = 0$ (reducing to $R_{\mathbf{G}} = 0$ in the port basis $C = I$); the identity $(L^{-1})^\sharp = (L^\sharp)^{-1}$ and the covariance $R' = URU^{-1}$ under simultaneous transformation of L , K , C are certified by the same exact symbolic harness as the rest of the paper.

Two structural corollaries are certified with them, and the first is critical for reading this paper. *The object matters*: the one-way closed form is a function of the single product $x = \bar{z}\bar{w}$, and the product commutes, so the scalar closed form is exchange-symmetric — R evaluated on it is identically zero, a false “reciprocal” verdict. Non-reciprocity lives on the complete two-point transfer kernel, the matrix with the reverse element vanishing ($G_{21} = f(\bar{z}\bar{w})$, $G_{12} = 0$), and only there is $R \neq 0$. This realizes a general projection blindness: in the port basis ($C = I$) the diagonal of R vanishes identically, and for a general symmetric pairing the certified statement is $v^T C (K - K^\sharp) v = 0$ for every probe v — no same-port (one-point) measurement can ever witness non-reciprocity; the invariant is irreducibly two-point. *Second*, under an external bias b the Onsager–Casimir relation $K_b = K_{-b}^\sharp$ (device against bias-reversed device) must not be conflated with reciprocity of the device against itself ($K_b = K_b^\sharp$): a gyrotropic kernel satisfies the former while violating the latter.

We record the following observations, verified symbolically on all cases above and on control kernels, organized by the two invariants. The diagonal (reproducing-kernel) sum of a single holomorphic sector, $-k \log(1 - z\bar{w})$, is exchange-Hermitian exactly — yet, as a two-point kernel, non-reciprocal ($R \neq 0$): it realizes the Hermitian-decoy quadrant, and earlier versions of this paper mislabeled it reciprocal; the Hermitian two-sector sum (a one-way half plus its mirror image) is exchange-Hermitian with the paired structure explicit; and the kernel $\log(z - \bar{w})$, which resembles two-dimensional chiral propagators, is exchange-Hermitian up to a branch constant ($\exp(\Delta) = -1$) and therefore lies on the Hermiticity axis up to that constant; its R is a separate question, not needed here. Unidirectional coupling, by contrast, deletes the mirror partner: the one-way kernels of Section 2.2 satisfy no exchange-Hermiticity relation, exactly because the reverse transfer element vanishes and nothing restores the conjugate-exchanged term. We summarize this as: the mirror partner marks exchange-Hermiticity, not reciprocity; the reciprocity statement is carried by $R = K - S(K)$ on the complete transfer kernel. One-wayness removes the partner and, independently, breaks R ; the unpaired anti-holomorphic half survives alone.

2.4 Classification of the kernel

The classification below concerns the scalar closed form (the content of the surviving element G_{21}); the reciprocity statement lives on the matrix kernel (Section 2.3). Three independent formula-level properties position the one-way kernel. First, cross-entanglement: the mixed logarithmic derivatives

$\partial_a \partial_b \log G_{21}$ across the two points, evaluated on all four conjugate pairings (z, w) , $(z, \bar{w})$, $(\bar{z}, w)$, $(\bar{z}, \bar{w})$, vanish except on $(\bar{z}, \bar{w})$ — the kernel is entangled between the two points, and specifically in the doubly-conjugated sector, as the winding bookkeeping dictates. (A single-pairing test is insufficient; we test all four.) Second, non-pairing: G_{21} is not invariant under the mirror map, and no mirror partner accompanies it, in contrast with the exchange-Hermitian controls above. Third, transcendence: for the physical ladder the closed form is a Lerch transcendent, while the flat-weight control is algebraic; transcendence is an independent axis supplied by the double-pole ladder. We do not attach any interpretation to this classification beyond the statements just made.

2.5 Certification methodology

All symbolic statements above — vanishing or non-vanishing of Wirtinger derivatives, the pairing tests, the exchange-Hermiticity axis and the independent reciprocity invariant R , including the branch-safe form — are certified by exact symbolic computation of Wirtinger derivatives (treating $z, \bar{z}, w, \bar{w}$ as independent variables), in a fixed verification harness that also runs the control battery (reproducing-kernel decoy, Hermitian paired sum, branch-constant case). Numerical evidence (series verification, finite-difference cross-derivatives on transcendental closed forms) supports but never replaces the symbolic certificates. We adopt this separation — a discoverer may propose, only the symbolic judge certifies — throughout the paper.

3 Boundary structure of the kernel

The closed form of Section 2.2 converges on the open unit polydisk $|x| < 1$, $x = \bar{z}_1 \bar{w}_2$, and the proposed protocol reads its boundary values. This section records the machine-verified structure of that boundary: its intrinsic and emergent character (Section 3.1), the chiral law of its crossing (Section 3.2), the interference relief along its tear (Section 3.3), and the exact log-scale calendar of the deep interference minima (Section 3.4). All results are structural characterizations of the kernel, verified in the repository accompanying this deposit; no physical claim is made.

3.1 An intrinsic, emergent boundary

Three structural facts about the boundary are worth recording. First, the boundary is intrinsic and dimensionless: it sits at the pure number $|x| = 1$, fixed by the geometry of the mode sum rather than by any parameter choice (the physical rates κ and δ enter the complex offset c , not the location of the boundary), and the formalism contains no dimensional constant from which any other distinguished scale could be built. Second, the boundary is emergent in the mode ladder: every truncation of the sum to N rungs is an entire function of x with no boundary at all, while the radius at which the truncated sum departs from the closed form approaches $|x| = 1$ monotonically as N grows (at the one-percent level, the departure radius is 0.904 at $N = 10$ and 0.991 at $N = 30$, and lies within 10^{-6} of the boundary beyond $N \simeq 100$). The boundary is therefore a property of the infinite ladder that no finite sub-ladder carries — the domain-level analogue of the fact, used throughout this section, that the one-way structure lives in the full ladder rather than in any single Jordan pair. Third, the closed form continues analytically beyond the boundary, where the defining series diverges: the kernel as a function outlives the kernel as a mode sum, at the price of a branch structure on $x \in (1, \infty)$ whose leading discontinuity is inherited from the dilogarithm ($2\pi i \ln x$; we verified the dilogarithm discontinuity numerically to nine digits, and state the kernel-specific jump at the level of the Lerch-to-dilogarithm reduction rather than as an independently certified result). We record these facts as a structural characterization only: they sharpen, but do not close, the boundary-limit question of Section 9, since a rigorous treatment of the $|x| \rightarrow 1$ limit in the presence of the complex offset remains open.

These characterizations were subsequently audited at the class level. The location of the boundary is dimensionless across coupling strengths (departure radius 0.990636 versus 0.987198 at $N = 30$ for

$\kappa = 0.2$ and $\kappa = 1.0$), and, across ladder families, the *existence* of the wall is generic while its *location* is set by the asymptotics of the mode weights: polynomially decaying weights place it at $|x| = 1$ for every tested power ($s = 1, 2, 3$, polylogarithm controls), while geometrically decaying weights of ratio r move it to $|x| = 1/r$ (machine-checked at $r = 0.5$). The wall is a property of the class; the choices pick the specimen.

3.2 The chiral phase law of the cut

The crossing of the boundary is *oriented*: writing the leading discontinuity of the closed form across the cut $x \in (1, \infty)$ as $\text{disc } \Phi(x, 2, a) = 2\pi i x^{-a} \ln x$, the complex offset $a = 1 + \kappa/(2i\delta)$ of the physical double-pole ladder — carried by the surviving sector alone, the one-way coupling having deleted the partner sector — makes the *phase* of the jump rotate along the cut (by $\arg x^{-(a-1)}$, about 9.2° from $x = 1$ to $x = 5$ at $\kappa/\delta = 0.2$), whereas the real-offset dilogarithm control has a frozen jump phase at every point of the cut. The ratio law $\text{disc}_{\text{one-way}}/\text{disc}_{\text{real-offset}} = x^{-(a-1)}$ was verified digit-for-digit against the dilogarithm control (itself machine-exact to 10^{-9}); the monodromy stacking across repeated crossings is arithmetic (equal sheet steps, ratios equal to unity to 10^{-21}). A published use of the cut-discontinuity phase law, read comparatively across the two sectors, as a (sufficient) reciprocity witness appears unreported in the literature we searched; the closest works we found diagnose directionality through spectral winding numbers or end-to-end Green's-function growth [25], through the phase of the drive [19], or through a frequency-dependent 2π phase gradient at a branch-cut crossing in non-Hermitian metasurfaces [24]; a position-dependent jump phase along the cut appears unreported in the literature we searched, and we flag the usual caveat that niche terminology may hide precedents.

Within the family studied here the control is the real-offset (scalar) one, so the frozen-versus-rotating dichotomy separates the two kernels. The rotation alone, however, is not a standalone reciprocity witness. Write $X = \bar{z}_1 \bar{w}_2$ and $Y = \bar{z}_2 \bar{w}_1$ for the two sector variables — the arguments of the two transfer configurations (G_{21} and G_{12} respectively, with $S(X) = Y$) — and D_X, D_Y for the cut discontinuities of the kernel in each sector variable, taken across the same ray. For a separable kernel $F_a(X) + F_a(Y)$ with the same complex offset in both sectors, each sector carries the same rotating jump phase while the kernel is exactly reciprocal ($R = 0$). The repaired witness is *comparative*: with $W(x) = D_X(x) - D_Y(x)$ the complex difference of the two sector jumps, $W \neq 0$ implies $R \neq 0$ (the witness is sufficient), while $W = 0$ is inconclusive — an entire antisymmetric addition to the kernel changes R without touching any cut (explicitly: $K_\varepsilon = \varepsilon(X - Y)$ is entire, has zero cut discontinuity, and carries $R = 2\varepsilon(X - Y) \neq 0$). The two sector cuts are compared with the same orientation, fixed once by the common parametrization of x ; the witness is stated in the port basis $C = I$, its certified domain — a general calibration rescales one sector relative to the other, so the raw difference requires a covariant reweighting, which we flag as an open certification target rather than assert here; physically, the two sector jumps correspond to the two calibrated transfer configurations (G_{21} and G_{12}). The complex difference is essential: a size asymmetry ($F_a(X) + \rho F_a(Y)$, $\rho > 0$ real, $\rho \neq 1$; a unimodular complex ρ would instead shift the angle component of the weight, not its modulus) leaves phase and slope identical in both sectors and is read only by the modulus of W , with the exact pointwise law $W(x) = (1 - \rho)D(x)$ — hence $\max_x |W| = |1 - \rho| \max_x |D(x)|$. All clauses (refusal on the reciprocal counterexample, firing on the one-way, size and speed columns, and the inconclusive limit) are certified by the same exact symbolic harness. The witness is established in the separable class $K = F(X) + G(Y)$ with two sectors; outside that class the sector jumps need not be separately defined, and no claim is made there.

The law was subsequently stress-tested across coupling strengths and ladder families. The ratio identity holds to machine precision (1.1×10^{-16} – 2.2×10^{-16}) at $\kappa = 0.05, 0.2$ and 1.0 , with the phase drift linear in the coupling, $\arg x^{-(a-1)} = (\kappa/2\delta) \ln x$ ($+0.99^\circ, +3.97^\circ, +19.86^\circ$ at $x = 2$ respectively); and the drift law is independent of the weight family (polynomially weighted ladders of every tested order obey the same law, with polylogarithm controls agreeing to better than 10^{-9} ; the symbolic identities are exact, the 10^{-9} figure is the numerical cross-check). Within the family studied the twist separates the one-way kernel from its real-offset control; it is not exclusive to the one-way class (a

paired reciprocal kernel with complex offset carries the same twist in both sectors, Section 3.2), and only the comparative difference witnesses R . The envelope belongs to the specimen.

3.3 Interference relief along the cut

When several kernels with different ladder spacings are summed, their tears stack on the same ray and the jumps add linearly, but the phase windings differ, and the resulting interference relief along the cut is governed by a *detuning* law: at fixed spread of ladder spacings, increasing the number of summed kernels averages the phases and flattens the relief, whereas at fixed number, widening the spread of spacings builds it (contrast growing by an order of magnitude and interior interference extrema appearing), while the real-offset (scalar) control remains flat at machine precision throughout. Together with the truncation behaviour of Section 3.1, this separates two emergence mechanisms living on the same object: the boundary itself emerges from the *number* of rungs, while structure *along* its tear emerges from their *diversity*. The emergence-with- N pattern of an analyticity boundary has well-known precedents in statistical mechanics — the accumulation of partition-function zeros in the thermodynamic limit [20] and the natural boundary of the 2D Ising susceptibility built from singularities that become dense as the order grows [21] — and we present the truncation ladder of Section 3.1 as an instance of that phenomenon in a two-point transfer kernel with complex offset, not as a new class.

The relief was further dissected with respect to the weights of the summed kernels. A *common* positive rescaling of all the weights is a pure volume knob: the normalized relief is exactly invariant. Relative changes of the positive moduli, by contrast, generally move the interference minima from $N \geq 3$ on — three needles with weights $(1, \alpha, 1)$ on windings $(0, 1, 2)$ have relief $|\alpha + 2 \cos u|$, whose deep minimum sits at $u = \arccos(-\alpha/2)$ for $0 < \alpha < 2$ and moves with α (beyond, the minimum locks at $u = \pi$); at $N = 2$ the minima positions are insensitive to the positive *moduli* (and a negative real weight carries an angle π , not a volume change at all). Relative complex phases between the weights move the pattern itself: for two needles a relative phase $\Delta\varphi = \varphi_2 - \varphi_1$ between the weights, with winding split $\Delta v = v_2 - v_1$, translates the interference minima by $-\Delta\varphi/\Delta v$ in logarithmic scale ($u = \ln x$; minima exactly at $u_n = ((2n+1)\pi - \Delta\varphi)/\Delta v$, Section 3.4), and for three or more needles it deforms the pattern — minima shift and their relative depths change — rather than acting as a volume knob; a phase common to all weights leaves the modulus relief invariant (the detailed window-dependence of the relative-phase effect was diagnosed and is recorded in the repository). Size disparity alone writes nothing: disparate weights on a single common winding speed give contrast at machine zero (9.6×10^{-16}), algebraically forced — the sum is a single fixed-length needle — so winding-speed disparity is the only source of relief. When sizes correlate with speeds the response is asymmetric: weighting the slow windings collapses the residual relief by an order of magnitude (a static-ballast effect: quasi-static needles drown the relative ripples), while weighting the fast ones raises it. Finally, as the number of summed kernels grows at fixed spread, the relief does not vanish: it saturates toward a distribution-dependent floor (near 0.048 for log-uniformly spaced ladders). The mechanism and the exact value of this floor admit a closed law and an elementary theorem, developed in Section 4.

3.4 The log-scale calendar of the deep rendezvous

The interference minima along the cut obey an exact and elementary law. With the trivial $1/x$ envelope removed, the normalized relief of a sum of N kernels equals, for all $x > 1$,

$$P(u) = \left| \sum_{k=1}^N w_k e^{i v_k u} \right|, \quad u = \ln x, \quad v_k = \frac{\kappa}{2\delta_k}, \quad (4)$$

an algebraic identity rather than an asymptotic statement: the relief is a finite sum of phasors rotating in *logarithmic* scale, almost-periodic in u by construction. The deep minima are therefore *log-periodic* rendezvous of the phasors — a form which is the classical signature of complex exponents in the

literature on discrete scale invariance [26, 27]. Machine panels: for $N = 2$ the relief is exactly periodic in u with period $T = 2\pi/|v_1 - v_2| = 94.247780$, matched to grid resolution; for the three-ladder configuration used above, four deep minima occur at $u = 10.00, 29.98, 49.93, 69.81$ (spacings ≈ 19.9 , near the extreme-winding beat), and the period law $T = 2\pi/|\Delta v|$ holds across four ladder spacings to 1.3×10^{-3} . This law also resolves an earlier provisional reading: interference structures reported at small x in intermediate work were partial descents toward the *first* rendezvous (located near $x \approx 2.2 \times 10^4$ in that configuration), outside every previously examined window; we record the correction rather than the provisional reading. The calendar is class-level in its *frequency content* — the beat frequencies are an algebraic identity in the v_k , independent of the weight family, with pairwise beat periods $T_{ij} = 2\pi/|v_i - v_j|$ between distinct windings ($v_i \neq v_j$) and a single global period only when the winding splits are commensurable (the two-needle law $T = 2\pi/|\Delta v|$ is the certified instance) — while the depth, selection and position of its deep minima are set by the weight moduli from $N \geq 3$ on and by the weight angles already at $N = 2$ (the relative-phase translation of Section 3.3); it is stable under fully randomized choirs of ladders — and it separates the kernel from its flat control: the real-offset control (all $v_k = 0$) is flat to machine zero, with no minima at all. Reciprocity by itself does not force $v_k = 0$ — a paired reciprocal kernel with rotating sectors carries a calendar too — so the calendar, like the twist, assigns one-wayness only through the comparative reading.

At the level of form, log-periodic oscillations are the hallmark of discrete scale invariance, where complex exponents arise from hierarchical structures or renormalization flows [27, 26]. Here the imaginary part of the exponent arises with no hierarchy anywhere in the construction — it is set by the complex offset of the damped double-pole ladder ($\kappa/2\delta$ per rung), the one-way coupling deleting the partner sector, though complex winding is not exclusive to one-way kernels (a paired reciprocal kernel with complex offset carries it too, Section 3.2); to our knowledge this hierarchy-free route to the motif is apparently new, and we flag the usual caveat that niche terminology may hide precedents.

4 The Fourier floor theorem

Section 3.3 ended on an empirical fact: as the number of summed kernels grows at fixed spread, the interference relief along the cut saturates toward a distribution-dependent floor. This section derives the exact law of that floor, proves it as an elementary theorem (the full proof is deposited in the repository accompanying this record), establishes the amplitude dichotomy of the continuum limit for the log-scale calendar of Section 3.4, and computes the floors of the canonical ladder families in closed form. Throughout, results labelled machine-verified were run against the harnesses in the repository; the theorem’s ingredients are classical, and are labelled as such.

4.1 The floor law

Encode a choir of N summed kernels by its *weighted winding measure* $\mu_N = (\sum_k w_k)^{-1} \sum_k w_k \delta_{v_k}$ — a probability measure when the weights are nonnegative, $w_k \geq 0$, which we assume for the main statement — on the winding speeds $v_k = \kappa/(2\delta_k)$ of Eq. (4). The normalized relief of Section 3.4 is then exactly the modulus of its Fourier transform, $P_N(u)/\sum_k w_k = |\hat{\mu}_N(u)|$ with $\hat{\mu}_N(u) = \int e^{i v u} d\mu_N(v)$, and the residual floor obeys: *as $N \rightarrow \infty$ at fixed distribution shape, on a fixed window in u , the relief converges to $|\hat{\mu}(u)|$ and the floor equals the window-contrast of the Fourier transform of the limiting weighted winding measure.* On the machine, ladder families already near their limit sit on the law (measured-to-computed contrast ratios 1.00–1.08 across the log-uniform, weighted and warped families tested), while families far from their limit approach it from above — which retroactively identifies the apparent “decay exponents” of the saturation curves as transients of this convergence, with no universal meaning. Two natural first-order candidate laws were refuted en route and are recorded for completeness: an incoherent (root-sum-square) ballast law fails with structured deviations up to a factor of ten — the floor is a *coherent* phenomenon — and a binary active/ballast criterion is degenerate on the

roughness window, where no phasor completes a full turn (maximal sweep ≈ 1.75 rad): the roughness floor lives entirely in the partial-rotation regime, while full turns belong to the calendar of Section 3.4.

4.2 An elementary theorem

Theorem (Fourier floor law). *Fix a window $I = [u_{\min}, u_{\max}] \subset (0, \infty)$ and let (μ_N) be probability measures on $[a, b] \subset [0, \infty)$ converging weakly to μ . Then $\hat{\mu}_N \rightarrow \hat{\mu}$ uniformly on I , and the window-contrast $\mathcal{C}(f) = (\max_I f - \min_I f) / (\max_I f + \min_I f)$ satisfies $\mathcal{C}(|\hat{\mu}_N|) \rightarrow \mathcal{C}(|\hat{\mu}|)$, the denominator being protected: $\max_I |\hat{\mu}| > 0$ always.*

Proof sketch. Pointwise convergence follows from weak convergence of measures against the bounded continuous family $v \mapsto e^{ivu}$; the uniform Lipschitz bound $|\hat{\mu}_N(u) - \hat{\mu}_N(u')| \leq b|u - u'|$ upgrades it to uniform convergence on the compact window; $\mathcal{C}$ is continuous in the sup norm wherever its denominator is positive; and the denominator is protected because $\hat{\mu}$ extends to an entire function of u with $\hat{\mu}(0) = 1$, so it cannot vanish on any interval. The full proof, together with a discrepancy lemma, is deposited in the repository. All ingredients are classical (equicontinuity, Paley–Wiener-type analyticity, discrepancy bounds); the contribution is the assembly fitted to the one-way tear. $\square$

The discrepancy lemma bounds the approach at rate $O(1/N)$ for quantile samplings of the limiting measure — a bound the machine shows is *not tight for all samplings*: midpoint quantile choirs converge at second order (measured log–log slope -2.00), endpoint choirs at first order (slope -1.02). The approach speed belongs to *how the choir samples its measure*, not to the measure alone; the naive first-order prediction was corrected by the machine before deposit, and the correction is part of the statement. Three corollaries close earlier threads. First, the real-offset (scalar) control is now a theorem line: all $v_k = 0$ means $\mu = \delta_0$, hence $\hat{\mu} \equiv 1$ and zero relief on every window — the machine-flat real-offset (scalar) controls of Sections 3.3 and 3.4 are instances of this one line. Second, uniform convergence makes the law estimator-agnostic: any functional continuous in the sup norm (contrast, r.m.s. ripple over mean, ...) converges alike, and the machine confirms both at ratio 1.000. Third, complex weights are covered by the same argument — again under weak convergence of the μ_N — provided $\sum_k w_k$ is bounded away from zero and every bound is read in total variation ($\|\mu_N\|_{\text{TV}}$ uniformly bounded), with μ_N then a finite complex measure rather than a probability measure and the normalized relief read as $P_N(u) / |\sum_k w_k| = |\hat{\mu}_N(u)|$ — the phase reweighting of Section 3.3 is a reweighting of μ .

4.3 The fate of the tides

The theorem’s sharpest consequence concerns the long-range amplitude of the relief carrying the calendar of Section 3.4, and splits on the nature of the measure. If μ is *finitely supported* — any finite choir — then $\hat{\mu}$ is a finite trigonometric sum, hence a Bohr almost-periodic function [28]: every value configuration recurs within any tolerance infinitely often — in particular a non-null finite choir cannot decay to zero — and the deep rendezvous recur forever (machine: for the three-ladder configuration, the maximal normalized relief equals 1.000 on the far windows $u \in [1000, 1200]$ and $u \in [3000, 3200]$ alike). If instead μ is *absolutely continuous* — a true continuum of windings — the Riemann–Lebesgue lemma gives $\hat{\mu}(u) \rightarrow 0$, with no universal rate in general; under mild regularity (a density of bounded variation on $[a, b]$, $a > 0$) the decay is $O(1/u)$, with constants set by the boundary values and the total variation of the density — for the log-uniform family the boundary term scales as $1/(ua)$: the maximal amplitude dies on far windows (machine, with a quantile proxy valid while $u \ll N$: maximal relief $0.0605 \rightarrow 0.0216$ between the two windows above, consistent with $1/u$). The dichotomy is therefore one of *amplitude*: a finite ladder’s relief cannot decay, while an absolutely continuous continuum’s amplitude dies away. Amplitude decay, however, does not erase the calendar: the persistence of zeros, the positions of the minima and the relative window contrast are decided by neither class alone — the uniform measure, whose closed form is computed in the next subsection, is absolutely continuous, yet its sinc transform keeps exact zeros at $u_n = 2\pi n/(b - a)$ forever, with window contrast equal to

one on every far window containing one (machine: contrast 0.9999999996 at $u \approx 1.9 \times 10^7$ while the maximal amplitude fell a thousandfold). Singular continuous measures form a third class, not treated here: their transforms need not decay (the middle-thirds Cantor measure persists on the subsequence $u_n = 2\pi 3^n$), and no universal destiny is claimed for them. No novelty is claimed for the ingredients, which are textbook; the dichotomy is recorded because it closes Sections 3.4 and 4.1 into one object.

4.4 Closed-form floors

Taking the theorem at its word, the floors of the canonical ladder families are computed rather than simulated. On the winding interval $[a, b] = [1/300, 1/3]$ used throughout (with $\Delta\text{Ci} + i\Delta\text{Si}$ denoting $\text{Ci}(bu) - \text{Ci}(au) + i[\text{Si}(bu) - \text{Si}(au)]$):

$$\hat{\mu}_{\log}(u) = \frac{\Delta\text{Ci} + i\Delta\text{Si}}{\ln(b/a)}, \quad \hat{\mu}_{1/v^2}(u) = \frac{\frac{e^{iau}}{a} - \frac{e^{ibu}}{b} + iu(\Delta\text{Ci} + i\Delta\text{Si})}{1/a - 1/b}, \quad (5)$$

$$\hat{\mu}_{\text{unif}}(u) = \frac{e^{ibu} - e^{iau}}{iu(b-a)}, \quad (6)$$

for the log-uniform, inverse-square (equivalently: linear-in- δ) and uniform measures respectively — the last being the cardinal sine. Their window-contrasts, 0.04764907, 0.00565563 and 0.06533857, match $N = 32768$ quantile choirs at relative accuracy 3.6×10^{-9} , 6.0×10^{-7} and 9.0×10^{-10} : the floors of the whole relief phenomenology are values of three formulas written with two-century-old functions. The derivation also exposed two identities hidden from the simulations: weighting a log-uniform ladder by $1/v$ (slow windings heavy, the static-ballast direction of Section 3.3) produces the effective measure dv/v^2 — *exactly* the linear-in- δ measure, so two independently measured floors (0.00568 and 0.00567) are one transform (closed value 0.00566); and weighting by v produces the uniform measure, so that floor is a contrast of the cardinal sine (0.06536 measured, 0.06534 closed). The compression chain of this section is then complete: the relief phenomenology of Section 3.3 reduces to three canonical measures, one law, one elementary theorem and three formulas — and the law can be read backwards, which is the subject of Section 5.

5 One-way spectroscopy of the surviving element

The floor law of Section 4 maps the composition of a choir to its relief. This section reads the map backwards: given only the measured relief along the cut, identify the family of the winding measure and recover its parameters. The result is an instrument that works — and that ships with its limits proven on the day of its construction, which we regard as part of the result.

5.1 The instrument

The inverter fits the closed-form dictionary of Eqs. (5)–(6) to a measured relief by least squares over the winding interval $[a, b]$, once per candidate family, and selects the family with the smallest residual. Blind tests against hidden finite choirs ($N = 64$, clean data) identify the log-uniform family correctly with a recovered to 1.34% and b to 0.11%, and the uniform family correctly with b to 0.21%; the inverse-square case, however, exposes a degeneracy of the short observation window: a log-uniform fit driven to its $a \rightarrow 0$ edge impersonates the $1/v^2$ family (residual 3.5×10^{-9} against 6.9×10^{-8} for the true family), and wins the selection. We record the failure verbatim; its diagnosis, and resolution, are the subject of the next subsection.

5.2 Proven limits

Three limits of the spectroscopy are established alongside it. First, the instrument is *exactly mirror-blind*: the reflected measure $v \mapsto a + b - v$ has transform $\hat{\mu}_{\text{mir}}(u) = e^{i(a+b)u} \overline{\hat{\mu}(u)}$, of identical modulus, so modulus-only data cannot distinguish a choir from its mirror (machine: maximal relief difference 5.7×10^{-15}). This is a hard bound of any modulus-only inversion, recorded rather than hidden. Second, *the window is the aperture*: information about a winding speed v lives at $u \sim 1/v$. On the short window the slow windings sweep only ≈ 0.02 rad and write essentially nothing — which at once explains the family degeneracy above (the candidate measures differ mainly at their slow edge) and makes the slow parameter fragile under noise (29% error on a at 1% relative noise, against 1.4% on b). Extending the window toward $u \sim 1/a$ (here $[0.05, 300]$, with $N = 1024$ so that the finite choir remains a valid proxy, $u \ll N$) dissolves both failures at once: the inverse-square family is then correctly identified with a thirty-fold residual separation and a recovered to 0.10%, and the noise-induced error on a collapses from 29.4% to 0.02%. Third, the instrument is *mute on the flat control by design*: a flat relief falls below the protocol contrast floor and the inversion is refused rather than hallucinated — the flat control is unreadable, as Corollary 1 of Section 4.2 demands.

5.3 Outlook for the measurement protocol

The spectroscopy upgrades the role of the boundary observables: a measurement of the relief along the cut, of the kind the recirculating-loop platform proposed below is designed to approach, would allow one not only to witness broken reciprocity (the comparative jump witness of Section 3.2: $W \neq 0 \Rightarrow R \neq 0$; strict unidirectionality $G_{12} = 0$ requires comparing the two calibrated transfer configurations) but to *read* the composition of the couplings that produce it — which modes, at which winding speeds, with what weights, within the proven limits above. The spectroscopy reads the weighted winding measure of the surviving element G_{21} ; it does not, by itself, measure $G_{21} - G_{12}$. Two acquisition-level caveats bound this reading. The witness W is a complex difference of cut discontinuities, while Section 5.2 proves modulus-only records mirror-blind: reading W , like any direct access to R_G , requires phase-referenced acquisition. Within the model, a known injected reference needle restores the missing phase: demodulating the relief against the reference beat, read at two reference phases in quadrature, recovers the complex transform — a phase-referenced augmentation of the intensity data, not a modulus-only inversion — lifting the mirror ambiguity of Section 5.2; the scheme, with a known injected reference needle $r e^{i(v_r u + \varphi_r)}$, and its exact demodulation law are certified in the repository accompanying this deposit across eight weight families, including size-disparate and size-speed-correlated choirs. We state this as an outlook on the mathematics of the kernel, not as an experimental claim.

6 Relation to prior structures

None of the ingredients used above is new in isolation; what we did not find a direct precedent for in our search is the combination of the forcing question, the mirror-deletion statement, the derived Lerch kernel, and the symbolic certification discipline. This section states, for each neighboring body of work, what is similar and what is structurally different. We make no priority claim beyond “no direct precedent found in our search”; we would welcome pointers to closer antecedents.

6.1 Bergman and Szegő reproducing kernels

Similar: reproducing kernels of holomorphic function spaces are two-point, cross-conjugate objects — functions of $z\bar{w}$ — and include logarithmic forms such as $-\log(1 - z\bar{w})$; at the formula level they mix one plain and one conjugated coordinate ($z\bar{w}$), a neighbouring cross-conjugate class to our doubly-conjugated sector ($\bar{z}\bar{w}$), and served as our principal controls. *Different*: reproducing kernels satisfy $K(\mathbf{r}_1, \mathbf{r}_2) = \overline{K(\mathbf{r}_2, \mathbf{r}_1)}$ by construction — they are exactly exchange-Hermitian. In the invariant table of

Section 2.3 this marks the Hermiticity axis $\Delta = 0$, not reciprocity: as a two-point kernel the Bergman form has $R \neq 0$, occupying the Hermitian-decoy quadrant. In our terms, they always come with their mirror partner. Our one-way kernel differs precisely in this respect: no exchange-Hermiticity relation holds, because the reverse transfer element vanishes identically. We regard the Bergman kernel as the sharpest decoy for the present construction: a transcendental, cross-conjugate formula that is not invariant under the mirror map alone and yet carries the mirror partner under the composed exchange $M \circ S$ ($\Delta = 0$). The exchange-Hermiticity test was introduced into our battery specifically because a formula-level gate without it accepts this decoy.

6.2 Cascaded quantum systems and reservoir-engineered non-reciprocity

Similar: the coupling mechanism we use is entirely standard. Unidirectional (cascaded) coupling of quantum systems goes back to Gardiner and to Carmichael [4, 5], and the reservoir-engineering recipe that makes one matrix element of a coupled pair vanish while keeping the other finite is that of Metelmann and Clerk [6], with several experimental realizations in superconducting circuits and optomechanics. *Different:* those works engineer and characterize the non-reciprocal coupling itself (isolation, directional amplification, scattering matrices). The added content here is downstream of the mechanism: the classification of the two-point transfer kernel that a whole ladder of such couplings produces (its conjugate-sector entanglement, its non-pairing, its transcendence class), and the statement that exchange-Hermiticity and reciprocity are independent kernel-level invariants, the one-way kernel violating both. We did not find this kernel-level independence table stated in the cascaded-systems literature covered by our search.

6.3 Mock modular forms and Appell–Lerch sums

Although the closed-form expression of the kernel involves the Lerch transcendent, reminiscent of certain mock-modular forms and Appell–Lerch sums, the object studied here is fundamentally different in both origin and structure. Our kernel arises from a unidirectional Lindblad coupling between counter-winding mode ladders, with a closed form convergent on the unit polydisk; it carries no modular transformation properties, and exhibits a strictly anti-holomorphic dependence on the product of conjugated coordinates. It should therefore not be regarded as a mock-modular form, but rather as an exchange-non-Hermitian one-way transfer kernel.

This is the most delicate comparison, and we treat it explicitly because the same special function appears on both sides. *Similar:* in the theory of mock theta functions as completed by Zwegers [7], and in its subsequent development [8, 9], a holomorphic q -series acquires good (modular) transformation behavior only after the addition of an explicitly non-holomorphic completion term; the building blocks are Appell–Lerch sums, which belong to the same Lerch family as the transcendent in our closed form. There is, moreover, a structural motif shared with our construction: an object that is “one half” of a symmetric pair, whose missing partner would restore a symmetry — modularity there, exchange-Hermiticity here — and whose interest lies precisely in the uncompleted half. *Different, and decisively so at the technical level:* (i) our kernel is a function of a single product variable $x = \bar{z}\bar{w}$ on the unit polydisk, with no modular variable τ , no lattice, no weight, and no transformation law — nothing in our construction transforms under $SL(2, \mathbb{Z})$ or any subgroup; (ii) the anti-holomorphic dependence of a mock form’s completion is arithmetic in origin (an Eichler-type period integral in $\bar{\tau}$), whereas ours is dynamical (winding bookkeeping of counter-propagating ladders under Lindblad-type one-way coupling); (iii) the parameter c in our Lerch transcendent is a dynamical complex offset, $\kappa/(2i\delta)$, not a lattice point; (iv) the symmetry whose breaking defines our object is the exchange of two spatial points, not a modular transformation. We therefore do not claim any correspondence between the two settings. Making one precise would require, at minimum, exhibiting a map under which the exchange operation realizes a modular transformation and the one-way kernel arises as the shadow or completion term of a specific mock object; we do not possess such a map, and we flag its existence or non-existence as

an open question (Section 9). The resemblance, as it stands, is a shared special-function family and a shared “missing-partner” motif — not an identity.

6.4 Non-reciprocal lattice models and the Hatano–Nelson asymmetry

Similar: unidirectional dynamics of Hatano–Nelson type — generated by an imaginary vector potential — is known to leave universal, closed-form fingerprints in its counting and echo statistics: in the interacting Hatano–Nelson model, the full counting statistics takes a universal Gaussian form whose mean scales with the imaginary vector potential [10], and the work statistics exhibits a universal high-energy tail with exponent -3 , originating from a temporal power-law decay of a generalized Loschmidt echo [11]. We read this as independent evidence that unidirectional dynamics writes universal structures into its generating functions. *Different:* the observables there are counting-statistics and Loschmidt-echo objects in a one-dimensional lattice, with no two-point spatial kernel, no conjugate-sector classification, and no exchange-Hermiticity statement; conversely, the Hatano–Nelson asymmetry itself is gauge-removable by an imaginary similarity transformation, whereas the one-way coupling used here is Lindbladian and not removable by any such similarity — a distinction that matters for the forcing question and that we test explicitly.

6.5 Exceptional points and double poles

Similar: the per-pair ingredient of our construction — a defective (Jordan) 2×2 block whose transfer function carries a double pole — is standard exceptional-point physics [12], and chiral exceptional points in whispering-gallery resonators are an active experimental subject. *Different:* a single exceptional point yields a single azimuthal Fourier tone of the two-point response and no transcendental structure; in our construction it is the coherent ladder of Jordan pairs, with its double-pole weights summed over the winding index, that produces the Lerch transcendent. We state this as a control result: the EP is necessary per pair but nowhere near sufficient; the ladder is the point.

Time-domain signatures of defective degeneracies also have an experimental precedent: Dietz *et al.* observed the algebraic-times-exponential decay (a t^2 intensity dependence) predicted at an exceptional point of a dissipative microwave billiard [13]. Relative to that precedent, the present proposal differs in the platform (a recirculating fiber loop read on an oscilloscope rather than a microwave cavity), in the observable set (the secular envelope is combined with a phase-swept revival comb and a single-versus-twin-comb witness, which the billiard setting does not probe), and in the object being certified (the kernel-level deletion of the exchange-Hermitian mirror partner by one-way coupling, certified symbolically upstream). The secular envelope itself is therefore not claimed as new; what appears new in our search is its use as a cheap, noise-robust certification channel for one-way kernels.

6.6 Non-Hermitian point-gap topology and directional amplification

The certification role proposed in Section 3.2 for the cut-discontinuity phase should be positioned against the standard witnesses of directionality in non-Hermitian lattice physics. There, non-reciprocity is diagnosed through the point-gap winding number of the determinant of the dynamic matrix as momentum traverses the Brillouin zone, in one-to-one correspondence with directional amplification and exponential end-to-end gain [29], and through the spatial growth of the bulk Green function, which the one-dimensional non-Hermitian winding number implies [30]; see also [25] for Green’s-function winding as a unifying invariant. These witnesses and ours share a family trait — an argument winding as the signature of one-wayness — but the objects differ: theirs wind in momentum space or grow in real space on lattice models, whereas ours is the position-dependent phase of the branch-cut discontinuity of a closed-form two-point kernel in the product of conjugated coordinates. We are not aware of an overlap of objects, and flag once more that niche terminology may hide precedents.

6.7 Discrete scale invariance and log-periodicity

The log-scale calendar of Section 3.4 places this work in contact with the literature on discrete scale invariance, where log-periodic corrections arise from complex critical exponents [26, 27] in systems with built-in hierarchies — hierarchical lattices, renormalization flows, and the broad range of applications surveyed in [27]. The present construction contains no hierarchy: the imaginary part of the exponent is set by the complex offset of the damped ladder, the one-way (Jordan) coupling deleting the partner sector — a route realized here, not the unique origin of complex winding. We therefore read the kernel as a hierarchy-free route to a classical motif, rather than as an instance of the classical mechanism — a positioning claim we keep hedged, as elsewhere in this paper.

6.8 Cousins of the forcing question

The question “when is anti-holomorphic dependence forced rather than chosen” has recognizable cousins in at least three settings: the holomorphic anomaly of topological string amplitudes [14], where consistency forces a controlled anti-holomorphic dependence on moduli; tt^* geometry [15], where holomorphic data are completed by a metric mixing the two sectors; and poly-analytic function theory [16, 17, 18], which studies function spaces genuinely dependent on $\bar{z}$, notably in higher Landau levels. *Similar*: in each case a structure that “would like” to be holomorphic is obliged, by consistency or by the physics, to carry anti-holomorphic content. *Different*: the objects (moduli-space amplitudes, ground-state metrics, single-particle function spaces), the mechanisms (anomaly equations, supersymmetric geometry, Landau-level structure), and the notion of removability all differ from ours; none of these settings poses the removability question for a two-point transfer kernel under non-reciprocal open-system dynamics, which is the specific question addressed here.

7 Proposed experimental platform and protocol

7.1 Chiral feedback bus on a recirculating fiber loop

We propose the following implementation. A recirculating fiber loop (length L , free spectral range $\delta = c/(n_g L)$) supports frequency-degenerate pairs of counter-propagating modes, indexed by the longitudinal number m ; on the loop, counter-propagation realizes the counter-winding pairing of Section 2 (the two members of a pair carry azimuthal factors $e^{+im\theta}$ and $e^{-im\theta}$). The loop is coupled to an add-drop bus; the clockwise output of the bus is routed through an optical isolator and re-injected onto the counter-clockwise input. This “chiral feedback bus” has three properties that match the mathematical construction of Section 2 without fine tuning. First, the two members of each pair are degenerate and couple to the same bus with the same rate, so the equal-diagonal condition of the per-pair Jordan block holds automatically. Second, the isolator suppresses the reverse path, so the vanishing of the reverse matrix element is a structural property of the setup rather than a tuned condition. Third, a single feedback loop serves the entire mode ladder simultaneously, with per-rung coupling phases $g_m = g_0 e^{im\phi}$ set by the loop delay ($\phi = \delta$ times the loop delay), which amounts to a rigid rotation of the ladder. Numerically, the ladder sum over this structure reproduces the closed form of Section 2.2 — evaluated on the unit circle of its convergence domain — to a relative accuracy of 1.4×10^{-7} in our implementation, which we regard as an identity check between the abstract kernel and the device model rather than as an approximation.

7.2 Mapped parameters

For concreteness we map the model onto catalogue-typical components (all values to be re-verified against the actual components of any experimental attempt): $L = 20$ m of standard single-mode fiber ($\delta/2\pi = 10.2$ MHz, round-trip time 98 ns); an in-loop isolator (insertion 0.8 dB, isolation 40 dB); an intracavity erbium amplifier (noise figure 5 dB) compensating the passive chain; a 70/30 output coupler;

a 1 GHz photoreceiver (about 98 rungs passed); a probe laser of 100 kHz linewidth; and a piezo fiber stretcher for the loop phase. These values give a loaded ratio $\kappa/\delta = 0.19$, an on-resonance conversion ratio of order unity, and an amplifier-noise-limited single-shot amplitude signal-to-noise ratio of order several hundred. One design tension is worth recording: an amplifier that fully compensates the passive losses raises the finesse and suppresses the frequency-domain visibility of the transcendental lineshape (which scales as $(\kappa/\delta)^2$); heavy output coupling restores κ/δ to the required range while the amplifier still compensates the passive chain. The time-domain protocol below is insensitive to this tension.

7.3 Time-domain protocol

Pulse the clockwise port; record the counter-clockwise impulse response. In the model, this response is a comb of revivals, one per round trip, carried by a secular envelope $t e^{-\kappa t/2}$ — the time-domain fingerprint of the double-pole (Jordan) structure, as opposed to the plain exponential of a diagonalizable pair. Three measurements are proposed. (i) Secular-envelope exponent: fit the revival peak heights to $t^p e^{-\kappa t/2}$; the Jordan structure predicts $p = 1$, a diagonalizable control predicts $p = 0$. (ii) Loop-phase sweep: the revival positions shift linearly with the piezo-controlled loop phase, with slope $1/\delta$; sweeping the phase at fixed ports scans the angular structure of the kernel without moving parts. (iii) Conditional reciprocity witness: within the loop model the one-way configuration predicts a single revival comb, and any reciprocal back-coupling produces a twin comb at the mirrored phase; a direct measurement of R_G additionally requires the reverse-injection acquisition (G_{12} : pulse the counter-clockwise port) alongside the forward one, with phase-referenced detection in both configurations (heterodyne or quadrature demodulation against a common reference, the calibration fixed once in the port basis, $C = I$): R_G is complex-valued, and modulus-only records are mirror-blind (Section 5.2). The phase-referenced branch is specified here as a requirement: its hardware realization (reference arm or local oscillator, I/Q or phase-stepped detection, relative calibration of the two directions) and its noise budget are beyond the simulated scope of this deposit and are flagged as future work. The twin-comb contrast is floored by the finite isolation of the isolator (energy floor $10^{-\text{isolation}/10}$); within the loop model the test therefore reads the modulus asymmetry implied by the mirror-deletion statement of Section 2.3 directly on an oscilloscope, down to that floor; the complex invariant R_G itself is not read on a modulus trace — its direct measurement requires the reverse-injection, phase-referenced acquisition of (iii). Figures 1 and 2 illustrate the two signatures on the deterministic model itself. We emphasize the epistemic order: these are proposed measurements; none has been performed.

8 Noise and imperfection analysis

8.1 Classification and grouped channels

For this protocol the decisive distinction is between within-record effects (acting inside one 0.8-microsecond record, hence able to distort the secular exponent) and between-shot effects (acting between averaged records, hence damping the revival comb: a shot-to-shot loop-phase spread σ suppresses rung m by $e^{-m^2 \sigma^2/2}$). Grouping ten physical effects by this criterion and by channel yields four families: (A) loop phase noise (thermal, acoustic, vibrational — the fiber thermo-optic response is of order 10^3 radians per kelvin over 20 m, so unstabilized drifts translate into radians over an averaging run); (B) intensity noise (amplifier relative intensity noise within the record; slow gain drift between shots); (C) polarization (state-of-polarization drift converting, through polarization-dependent loss, into slow amplitude fading); (D) parasitic reflections and finite isolation (a genuine weak twin comb at the isolation floor; distributed Rayleigh backscatter sits well below it for a 20 m loop). Kerr nonlinearity and third-order dispersion are included as switches and shown negligible at milliwatt powers (of order 10^{-5} radian per round trip, and 10^{-3} of the second-order dispersion, respectively). All magnitudes are literature-typical and flagged as such.

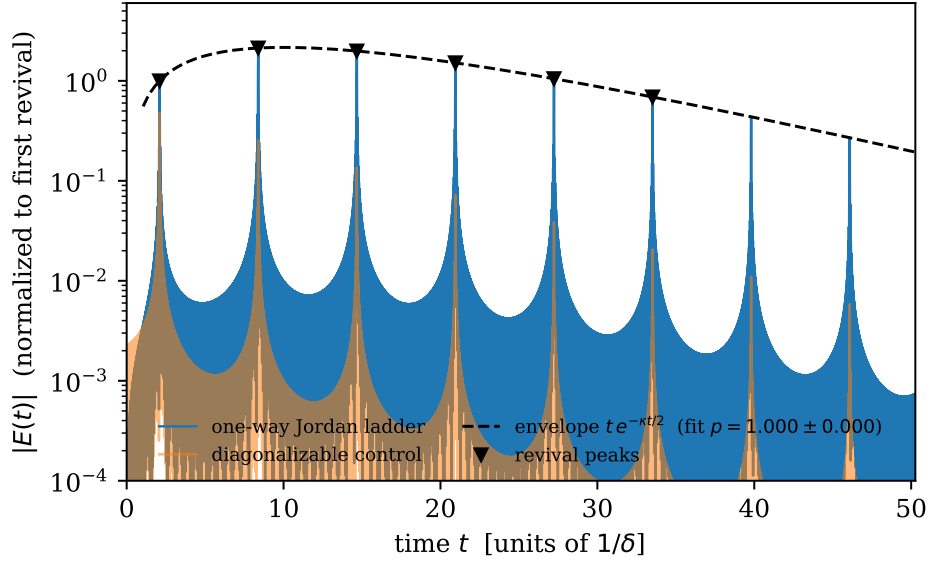


Figure 1: Simulation of the model of Section 2 in the mapped dimensionless regime ($\kappa/\delta = 0.2$, $M = 300$ rungs, loop phase $\phi = 2.1$): revival comb $|E(t)|$ of the one-way Jordan ladder versus the diagonalizable control, with the secular envelope $t e^{-\kappa t/2}$ anchored on the fitted revival peaks. The three-regressor fit of Section 7.3 recovers $p = 1.0001 \pm 0.0001$ (Jordan prediction $p = 1$) and $\hat{\kappa} = 0.2000$. Deterministic model output; no experimental data.

8.2 Simulation results

With additive detection noise alone, the secular exponent separates from the diagonalizable value at the ten-sigma level in a single shot at amplitude signal-to-noise 10, and at about forty sigma at signal-to-noise 3 with one hundred oscilloscope averages. Replacing white noise by amplifier noise recirculating on the loop resonances — the least favorable spectral shape, since the noise then peaks exactly where the signal lives — degrades this only mildly. The exponent is likewise unaffected by detector-bandwidth truncation down to about fifteen rungs, by slow polarization fading, by gain jitter, and, notably, by shot-to-shot loop-phase jitter: the secular envelope of the revival peaks does not depend on the ladder size, so the Jordan certificate survives phase noise that visibly destroys the comb. What phase noise does destroy is the many-rung content of the comb — the part of the signal that carries the transcendental class — with an effective ladder m_{eff} of order $1/\sigma$. Loop dispersion is subcritical at the mapped design and produces a clear failure signature (late revivals broadening with revival index) only when exaggerated far beyond it. Finite laser linewidth acts, after ensemble averaging, as an additional decay rate π times the linewidth: it biases the extracted κ but provably not the exponent, whose regressor is separated in the fit. Finally, two facts matter for the conditional reciprocity witness: the isolation floor of a standard 40 dB isolator lies below the realistic noise-floor of the twin-comb contrast, so single-stage isolation suffices initially; and loop phase noise pollutes the twin-comb contrast roughly tenfold before it kills the comb, so the witness demands a phase-quiet loop even more than the comb itself does. The analysis above covers the modulus observables; the phase-referenced acquisition required for a direct measurement of R_G carries its own error terms (reference drift, I/Q imbalance, calibration error) which are not simulated here.

8.3 Scope and ranking

Within the explored window of literature-typical and stressed values, loop phase stability dominates every other imperfection; intensity, polarization and reflection channels are benign. We state the ranking

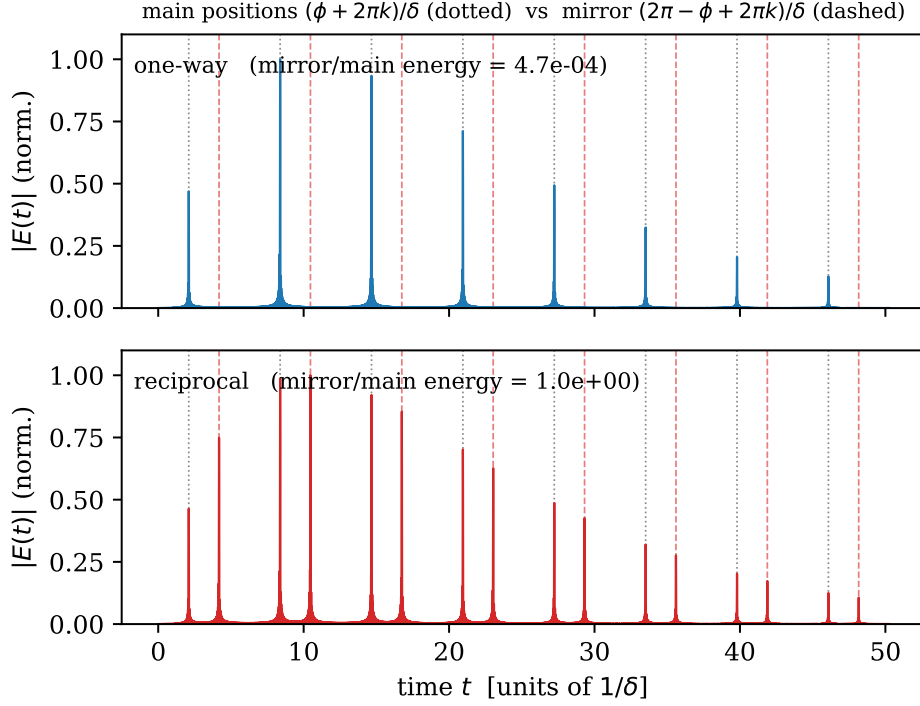


Figure 2: The mirror-deletion law in the time domain (same model and parameters as Fig. 1). Top: one-way configuration — a single revival comb at the main positions $(\phi + 2\pi k)/\delta$ (dotted); the energy at the mirrored positions $(2\pi - \phi + 2\pi k)/\delta$ (dashed) is at the numerical floor (mirror-to-main energy ratio 4.7×10^{-4}). Bottom: matrix reciprocal control (G_{12} restored) — the twin comb appears at the mirrored positions with ratio ≈ 1 . Deterministic model output; no experimental data.

with its exact scope: the damage mechanisms and their scalings in the mode index are established in the model; the ordering for a specific real device depends on the relative magnitudes of its actual imperfections and remains to be established from measured component data. The corresponding design guidance — protect the loop phase first — is conditional on that window.

9 Discussion and limitations

The additions of this version call for their own perimeter statement. Sections 3–5 are structural characterizations of the kernel as a mathematical object: machine-verified simulation throughout, one elementary hand proof whose ingredients are classical and labelled as such, and closed forms faced against the simulations to their last digits. None of it adds a physical claim. The log-periodic form is classical in the discrete-scale-invariance literature (Section 6.7); what we read as apparently new — the hierarchy-free one-way route to it, and the jump-phase witness — is stated with the hedge it deserves, and its arbitration belongs to external review, not to this deposit.

9.1 Modeling assumptions

Four idealizations deserve explicit statement. The mode families are taken as ideal counter-winding ladders with a strictly linear frequency ladder; real devices have dispersion, which we treat perturbatively in Section 8 but which deforms the exact closed form. The one-way coupling is Lindbladian and exact (the reverse element vanishes identically); a physical isolator has finite isolation, which converts the exact statement “the mirror partner is deleted” into the quantitative statement “the mirror partner is

suppressed to the isolation floor”. The laser is treated at the ensemble level, where its finite linewidth becomes a deterministic additional decay; individual records carry phase-diffusion realizations that the ensemble treatment averages. Finally, every numerical value in Sections 7–8 is literature- or catalogue-typical; none is a measurement of an actual component.

9.2 What a measurement would and would not establish

Observation of the three signatures — secular exponent, phase-swept revival comb, single-versus-twin comb down to the isolation floor — would establish the Jordan structure, the anti-holomorphic sector structure of the surviving element (Section 2), and the model-level comb suppression; establishing strict unidirectionality $G_{12} = 0$ additionally requires the reverse-injection configuration of Section 7.3. It would not establish anything beyond that. In particular it would carry no implication for arithmetic questions (Section 6.3), no statement about systems other than the one measured, and no foundational or cosmological consequence. Conversely, the measurement is informative in failure: each signature refutes a specific structural claim, as follows.

9.3 Falsifiability

The proposal is refutable in parts. (i) If the secular envelope shows an exponent consistent with zero under conditions where per-pair degeneracy and one-way coupling are independently verified, the Jordan-ladder structure is refuted. (ii) If the conversion channel of the two-point response depends on the coordinate difference rather than the coordinate sum, the one-way kernel structure is refuted. (iii) If a twin comb appears in the one-way configuration at a contrast far above the independently measured isolation floor, the mirror-deletion statement is refuted. (iv) The converse direction is bounded: none of the three signatures, if observed, establishes more than the kernel structure itself.

9.4 Open questions

Three questions are open at the level of mathematics. First, the exact status of the resemblance to mock-modular completions: whether any map exists under which the exchange operation realizes a modular transformation and the one-way kernel arises as the shadow of a specific mock object; we possess no such map and no argument excluding one. Second, the completeness of the classification: our three formula-level properties separate all cases we have constructed, but we have no theorem that they separate all kernels of physical origin. Third, the boundary behavior: the closed form lives on the open unit polydisk and the proposed measurement reads its boundary values; a rigorous treatment of the boundary limit, in the presence of the complex ladder offset, would strengthen Section 7. A structural characterization of this boundary — its intrinsic, dimensionless location, its emergence under truncation of the mode ladder, and the analytic continuation and branch structure beyond it — is given in Section 3; the oriented (chiral) crossing law and the detuning-governed interference relief along the cut are characterized there as well; the rigorous limit statement remains open.

10 Conclusion

We have posed the removability question for anti-holomorphic structure in the setting where, we argue, it is best posed — the two-point transfer kernel of a non-reciprocal open system — and answered it at the level of certified formula structure: one-way coupling between counter-winding mode ladders produces a Lerch-transcendent kernel whose exchange-Hermitian mirror partner is absent and whose reciprocity invariant $R_G = G - S(G)$, evaluated on the complete two-point transfer kernel (the scalar closed form alone is exchange-symmetric and blind to it), is non-zero — two independent violations that are both exhibited by the one-way construction at formula level (they follow from $G_{12} = 0$ with nonvanishing transfer; they do not by themselves characterize strict unidirectionality). We have proposed a fiber-loop

measurement of this structure, with an analysis of the realistic imperfections of its modulus observables (the phase-referenced branch required for R_G is specified but not simulated) and explicit criteria under which the proposal would be refuted. No experimental data exist; the next step is a measurement.

Code and data availability

The symbolic certification harness, the simulation scripts, and the findings files underlying every certified claim in this paper are archived alongside this deposit. This work extends the author’s eml* program on the detection of anti-holomorphic dynamics; the corresponding paper and software are archived at Zenodo under DOIs [10.5281/zenodo.20272784](https://doi.org/10.5281/zenodo.20272784) (paper) and [10.5281/zenodo.20273262](https://doi.org/10.5281/zenodo.20273262) (software), to which the present deposit is linked.

AI assistance disclosure

The author, an independent researcher, developed this work with AI assistants used as coding, auditing, and drafting tools under a strict verification discipline: every symbolic claim was certified by exact Wirtinger-derivative computation executed on the author’s machine, and no numerical result reported here was produced or accepted without such execution. Responsibility for the content rests with the author.

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